Yalamanchili Harsha Vardhan
Computer Science & Engineering
Indian Institute of Technology Bombay

24B1069
B.Tech.
Gender: Male
DOB: 03/01/2007

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2028	9 **Core:** 62 **| Total:** 62
Intermediate	CBSE	Narayana English Medium School	2024	97.80%
Matriculation	CBSE	Narayana High School	2022	97.60%

Scholastic Achievements

- Attained **All India Rank 170** in the **Joint Entrance Examination Advanced**, among **180,000+** aspirants *2024*
- Secured **Rank 84** in **TS EAPCET**, a premier state-level entrance exam, among **254,000+** contenders *2024*
- Achieved **All India Rank 267** in the **Joint Entrance Examination Main**, among **1,400,000+** aspirants *2024*
- Secured **Rank 119** in **AP EAPCET**, a major state-level entrance test, out of **339,000+** participants *2024*

Key Projects

Summer Of Quant | *Quant Community, IIT Bombay* *(June - July 2025)*

- Studied **Markov chains**, **stochastic** and **Poisson processes** with applications in quantitative modeling
- Gained exposure to **time series analysis**, **feature engineering**, and signal generation for trading strategies
- Learned **quantitative trading strategy design**, focusing on risk management, **backtesting**, and model robustness
- Secured a **Certificate of Excellence** and ranked among the top 10 performers in the Summer of Quant program

SAT-Based Sokoban Solver | *Course Project: Logic in Computer Science* *(Aug 2025)*

Instructor: Prof. S. Krishna

- Designed a **robust multi-dimensional state-space encoding** using **PySAT**, translating Sokoban game mechanics into **Boolean satisfiability** with 6+ constraint types, including movement logic and non-overlap conditions
- Enhanced solver performance using **unit propagation** and **clause learning**, while optimizing variable encoding to efficiently handle complex **SAT** instances and enable faster solution discovery and more scalable solutions
- Constructed **CNF** clauses modeling **collision dynamics** and **transition logic**, enabling valid solution paths

Stock Price Prediction Using LSTM | *Self Project* *(Dec 2024)*

- Built a stock price prediction model using **LSTM networks** to capture temporal dependencies in past market data
- Collected, cleaned, and visualized stock data with **Pandas, NumPy, Matplotlib, yfinance**, and **Scikit-learn**
- Implemented real-time stock data collection and performed advanced **feature engineering** with **yfinance**

Applied AI Learning Project | *Alcheringa, IIT Guwahati* *(Jan - Feb 2025)*

- Completed an **8-week AI program**, gaining hands-on experience with modern AI and machine learning techniques
- Gained strong expertise in **machine learning algorithms, data processing, and predictive modeling**, and applied these advanced technical skills effectively to analyze data, build models, and solve real-world problems
- Demonstrated a strong and consistent commitment to continuous learning and advanced technical skill development

Restaurant E-Commerce Website | *Web Development Project* *(Jan - Feb 2025)*

Devcom, IIT Bombay

- Built a full-stack **restaurant web app** with an interactive digital menu and real-time dynamic cart system
- Developed a fully responsive frontend in **React JS** and backend in **Node.js**, gaining hands-on experience in **modular web development**, efficient state management, and scalable e-commerce application design

AI Chess Engine | *Self Project* *(June - July 2025)*

- Understood several core concepts of **Game Theory** such as **MinMax Theorem**, **Nash Equilibrium** and **Zero Sum Games** to develop optimal strategies for a wide variety of competitive scenarios and decision-making problems
- Designed an **Advanced Evaluation Function** for **Chess Analysis** incorporating current positional insight and future position forecast to make a **chess bot** capable of solving various **puzzles** with **90%+ accuracy** efficiently
- Extended chess bot capabilities to play complete matches, integrating features like **Opening Book Lookup**, **MinMax Theorem**, and **Alpha-Beta Pruning Optimization**, successfully defeating a **1500-rated** Lichess bot

Technical Skills

Languages	C/C++, Python, Bash, Assembly, Verilog, HTML, CSS, JavaScript, SQL, R, Java
Software & Tools	Git, GitHub, Docker, Make, LaTeX, Markdown, Sed, Awk, Jupyter, ReactJS, Fusion360
Crypto/Security	PyCryptodome, Hash Functions, Side-Channel Analysis, Constant-Time Programming
Data/ML Libraries	NumPy, Pandas, MatPlotLib, SciPy, TensorFlow, PyTorch, OpenCV, Scikit-learn
Embedded Systems	Arduino, FPGA (Verilog), Circuit Design, PCB Design, IMU/Gyroscope Sensors